

33. Drag'n'Drop (Mouse Events se)

Browser ke apne native drag events (`dragstart`, `dragend`) hain, par unki limits hain (kisi area se drag rokna, sirf horizontal/vertical drag, mobile support kamzor). Isliye hum mouse events se khud banate hain.

Basic algorithm

1. `mousedown`: element ko move ke liye taiyar karo (`position: absolute`, `z-index` upar).
2. `mousemove`: `left/top` badlo.
3. `mouseup`: kaam khatam karo, handlers hatao.

js

```
ball.onmousedown = function(event) {
  ball.style.position = 'absolute';
  ball.style.zIndex = 1000;
  document.body.append(ball); // body me daalo taaki body ke hisaab se positi

  function moveAt(pageX, pageY) {
    ball.style.left = pageX - ball.offsetWidth / 2 + 'px';
    ball.style.top = pageY - ball.offsetHeight / 2 + 'px';
  }
  moveAt(event.pageX, event.pageY);

  function onMouseMove(event) { moveAt(event.pageX, event.pageY); }
  document.addEventListener('mousemove', onMouseMove);

  ball.onmouseup = function() {
    document.removeEventListener('mousemove', onMouseMove);
    ball.onmouseup = null;
  };
};
```

2 zaruri baatein

1. **Browser ka native drag band karo**, warna ball "fork" ho jaata hai (clone drag hota hai):

js

```
ball.ondragstart = function() { return false; };
```

2. `mousemove` `document` par sunte hain, ball par nahi, kyunki tez move me pointer ball se bahar kahin bhi jump kar sakta hai.

Sahi positioning (shift yaad rakho)

Bina iske, ball ke kinare se pakdo to wo jump karke center me aa jaata hai. Isliye `mousedown` par pointer aur ball ke top-left ka farak yaad rakho:

```
js

let shiftX = event.clientX - ball.getBoundingClientRect().left;
let shiftY = event.clientY - ball.getBoundingClientRect().top;

// mousemove me
ball.style.left = event.pageX - shiftX + 'px';
ball.style.top  = event.pageY - shiftY + 'px';
```

Droppable targets (kis par gira rahe hain?)

Seedhi soch: droppable elements par `mouseover/mouseup` lagao. **Ye kaam nahi karta**, kyunki dragging element hamesha sabse upar hota hai aur events sirf top element par hote hain.

Solution: `document.elementFromPoint(clientX, clientY)`. Isse pehle dragging element ko chhupao (`hidden = true`), warna wahi wapas milega:

```
js

let currentDroppable = null;

function onMouseMove(event) {
  moveAt(event.pageX, event.pageY);

  ball.hidden = true;
  let elemBelow = document.elementFromPoint(event.clientX, event.clientY);
  ball.hidden = false;

  if (!elemBelow) return; // window ke bahar to null

  let droppableBelow = elemBelow.closest('.droppable');

  if (currentDroppable !== droppableBelow) {
    if (currentDroppable) leaveDroppable(currentDroppable);
    currentDroppable = droppableBelow;
    if (currentDroppable) enterDroppable(currentDroppable);
  }
}
```

Summary

- Flow: `ball.mousedown` -> `document.mousemove` -> `ball.mouseup` (aur `ondragstart` cancel karo).
- Shift (`shiftX/Y`) yaad rakho.
- Draggable ka pata `elementFromPoint` se lagao.
- Isi base par aur features: `mouseup` par drop finalize, highlight, area/direction limit, aur **event delegation** se hazaron elements ka drag manage karna.

Tasks (short)

- **Slider:** sirf horizontal drag, thumb ko slider ki width ke andar rokna ($\text{left} < 0$ ya $> \text{right}$ edge par clamp).
- **Superheroes drag karna:** ek `document` handler (delegation), dragging ke waqt `position:fixed` (coordinates aasan), end me `absolute`; window ke top/bottom par pahunchne to `window.scrollTo` se page scroll.